Assignment 03

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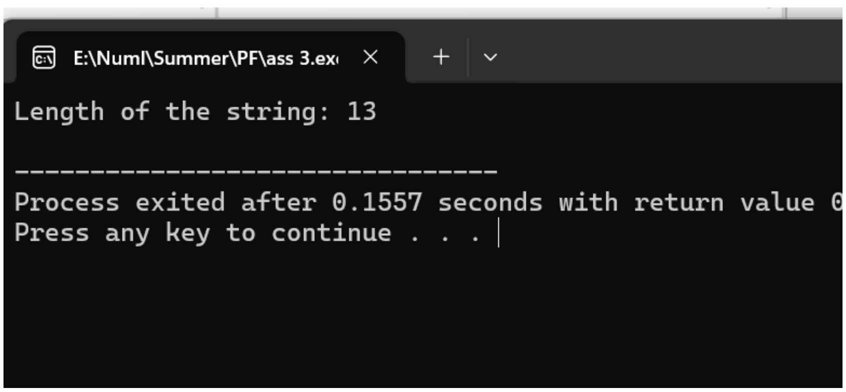
`length()`: Returns the length of the string

```
#include <iostream>

#include <string>

using namespace std;


int main() {

   string str = "Hello, World!";

   cout << "Length of the string: " << str.length() << endl;

   return 0;

}
```



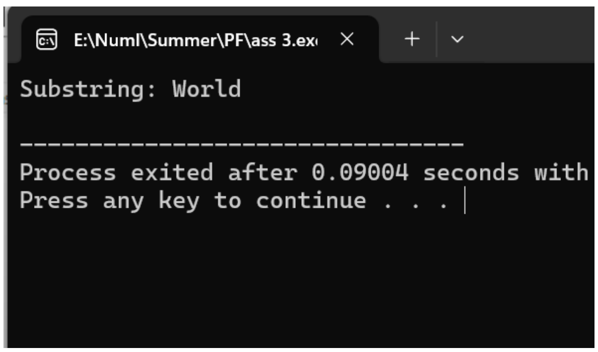
Explanation:

- The `length()` function returns the number of characters in the string `str`.
- For `"Hello, World!"`, the length is `13`.

`substr()`: Returns a substring from a string.

```
#include <iostream>

#include <string>

using namespace std;
```

```
int main() {
   string str = "Hello, World!";
   string sub = str.substr(7, 5);
   cout << "Substring: " << sub << endl;
   return 0;
}
```



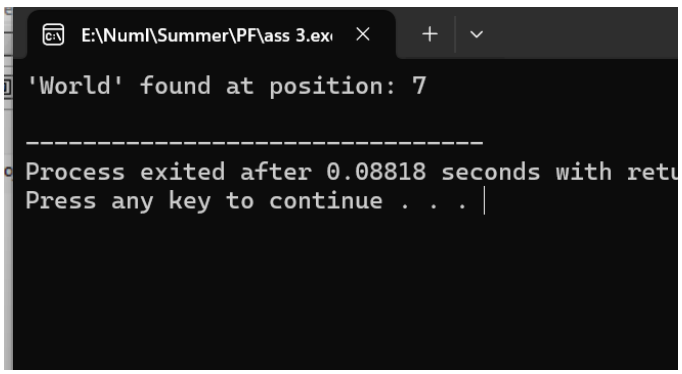
Explanation:

- The `substr(pos, len)` function returns a substring starting at position `pos` and of length `len`.
- Here, it starts at index `7` and returns the next `5` characters, resulting in `"World"`.

`find()`: Finds the first occurrence of a substring.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str = "Hello, World!";
```

```
    size_t pos = str.find("World");

    if (pos != string::npos)

        cout << "'World' found at position: " << pos << endl;

    else

        cout << "'World' not found" << endl;

    return 0;

}
```



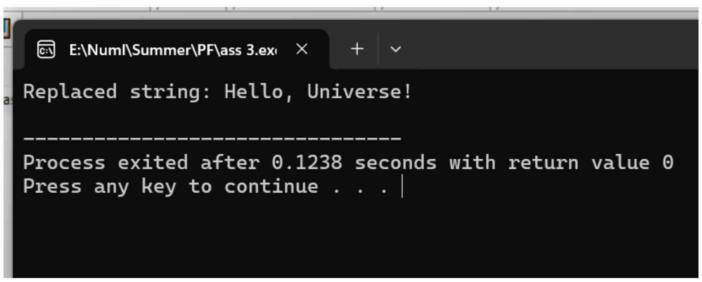
Explanation:

- The `find()` function searches for the first occurrence of the substring `"World"`.
- It returns the position of the first character of the substring.

`replace()`: Replaces part of a string with another string.

```
#include <iostream>

#include <string>
```

```
using namespace std;


int main() {

    string str = "Hello, World!";

    str.replace(7, 5, "Universe");

    cout << "Replaced string: " << str << endl;

    return 0;

}
```



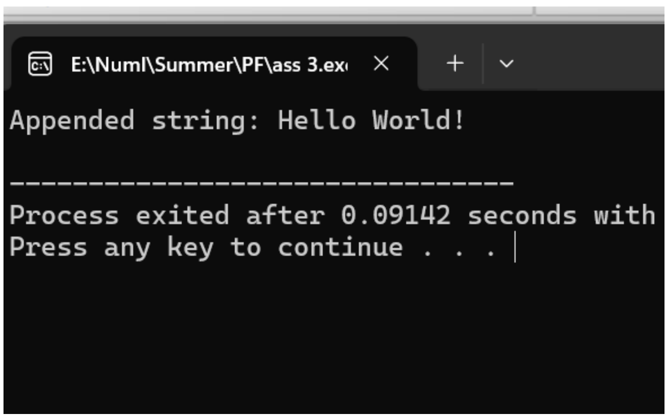
Explanation:

- The `replace(pos, len, new_str)` function replaces the part of the string starting at `pos` and of length `len` with `new_str`.
- Here, `"World"` is replaced with `"Universe"`.

`append()`: Appends a string to another string.

```
#include <iostream>

#include <string>

using namespace std;
```

```
int main() {
    string str1 = "Hello";
    string str2 = " World!";
    str1.append(str2);
    cout << "Appended string: " << str1 << endl;
    return 0;
}
```



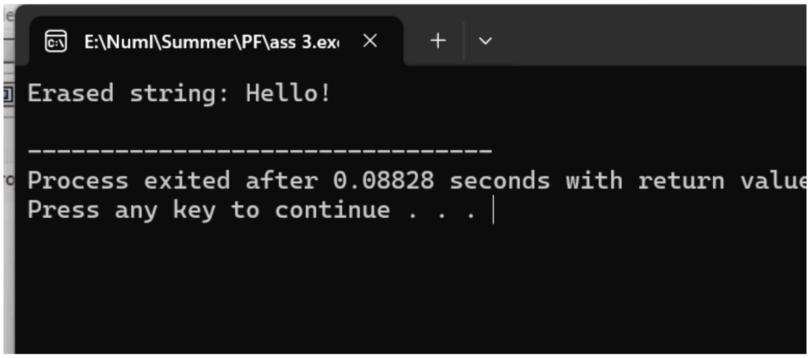
Explanation:

- The `append()` function adds `str2` to the end of `str1`.

`erase()`: Erases part of a string.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str = "Hello, World!";
```

```
    str.erase(5, 7);

    cout << "Erased string: " << str << endl;

    return 0;

}
```

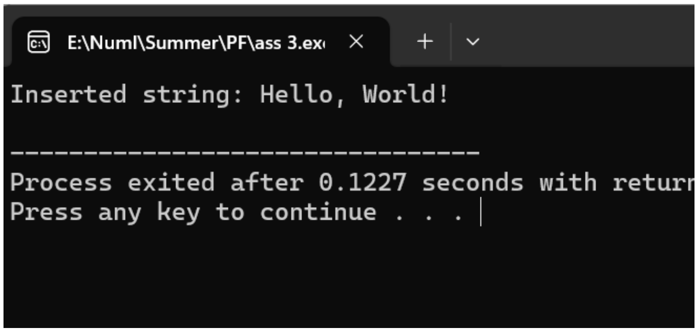


Explanation:

- The `erase(pos, len)` function removes `len` characters from the string starting at `pos`.
- Here, `" World!"` is removed.

`insert()`: Inserts a string into another string.

```
#include <iostream>

#include <string>

using namespace std;


int main() {

    string str = "Hello,!";

    str.insert(6, " World");

    cout << "Inserted string: " << str << endl;

    return 0;

}
```

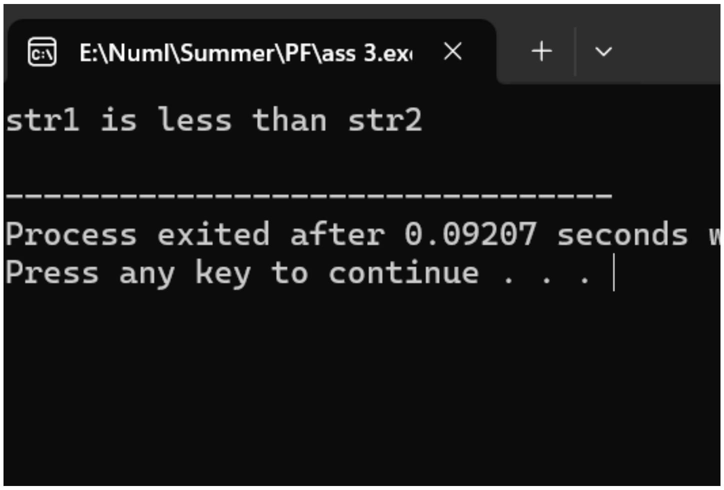


Explanation:

- The `insert(pos, str)` function inserts `str` at position `pos` in the string.
- `" World"` is inserted after `"Hello,"`.

`compare()`: Compares two strings.

```
#include <iostream>

#include <string>

using namespace std;


int main() {

   string str1 = "Hello";

   string str2 = "World";

   int result = str1.compare(str2);

   if (result == 0)

      cout << "Strings are equal" << endl;

   else if (result < 0)

      cout << "str1 is less than str2" << endl;

   else

      cout << "str1 is greater than str2" << endl;

   return 0;

}
```

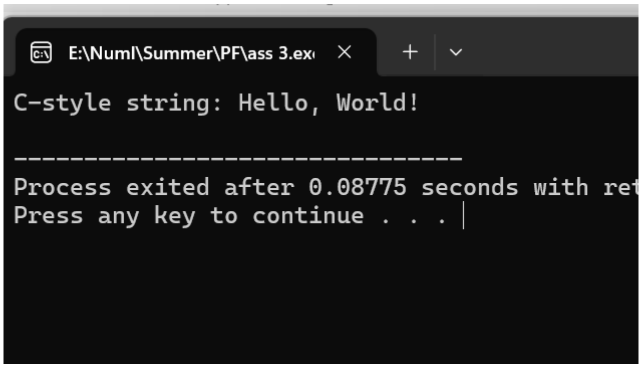


Explanation:

- The `compare()` function compares two strings lexicographically.
- It returns `0` if they are equal, a negative value if `str1` is less than `str2`, and a positive value if `str1` is greater.

`c_str()`: Returns a C-style string (null-terminated).

```
#include <iostream>

#include <string>

using namespace std;


int main() {

   string str = "Hello, World!";

   const char* cstr = str.c_str();

   cout << "C-style string: " << cstr << endl;

   return 0;

}
```

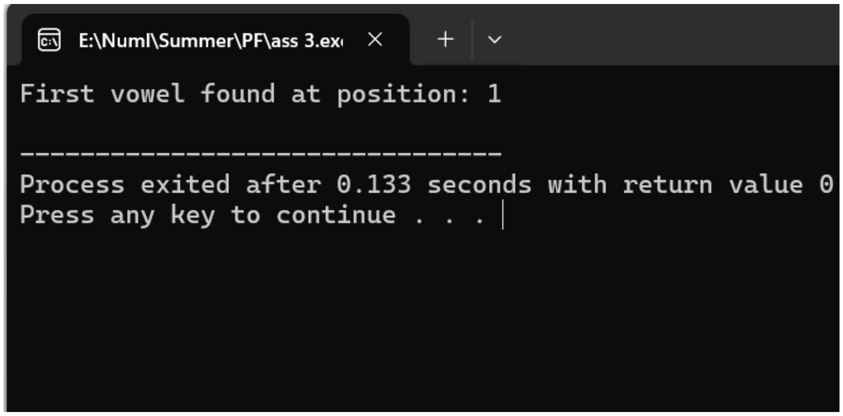


Explanation:

- The `c_str()` function returns a pointer to a null-terminated C-style string.

`find_first_of():` Finds the first occurrence of any character from a set.

```
#include <iostream>

#include <string>

using namespace std;


int main() {

   string str = "Hello, World!";

   size_t pos = str.find_first_of("aeiou");

   if (pos != string::npos)

      cout << "First vowel found at position: " << pos << endl;

   else

      cout << "No vowel found" << endl;

   return 0;

}
```



Explanation:

- This code finds the position of the first vowel (`a, e, i, o, u`) in the string and prints its position.
- The first vowel in `"Hello, World!"` is `e`, found at position `1`.